Chain-of-Thought Reasoning without Prompting

Xuezhi Wang[1] **and Denny Zhou**[1]
[1]Google DeepMind, [1]{xuezhiw, dennyzhou}@google.com

In enhancing the reasoning capabilities of large language models (LLMs), prior research primarily focuses on specific prompting techniques such as few-shot or zero-shot chain-of-thought (CoT) prompting. These methods, while effective, often involve manually intensive prompt engineering. Our study takes a novel approach by asking: Can LLMs reason effectively without prompting? Our findings reveal that, intriguingly, CoT reasoning paths can be elicited from pre-trained LLMs by simply altering the *decoding* process. Rather than conventional greedy decoding, we investigate the top-k alternative tokens, uncovering that CoT paths are frequently inherent in these sequences. This approach not only bypasses the confounders of prompting but also allows us to assess the LLMs' *intrinsic* reasoning abilities. Moreover, we observe that the presence of a CoT in the decoding path correlates with a higher confidence in the model's decoded answer. This confidence metric effectively differentiates between CoT and non-CoT paths. Extensive empirical studies on various reasoning benchmarks show that the proposed CoT-decoding effectively elicits reasoning capabilities from language models, which were previously obscured by standard greedy decoding.

arXiv:2402.10200v2 [cs.CL] 23 May 2024

1. Introduction

Large language models (LLMs) have demonstrated remarkable performance on various complicated reasoning benchmarks (Anil et al., 2023; Brown et al., 2020; Chowdhery et al., 2023; Gemini, 2023; OpenAI, 2023; Romera-Paredes et al., 2023). These reasoning capabilities of LLMs are typically elicited by prompting techniques (Brown et al., 2020), which can be few-shot prompting with intermediate steps augmented demonstration exemplars (Chen et al., 2023b; Gao et al., 2022; Nye et al., 2021; Wei et al., 2022; Yao et al., 2023; Zhou et al., 2023a), or zero-shot prompting with specific instructions which ask for showing certain intermediate steps (Kojima et al., 2022; Yasunaga et al., 2023). The other prevalent strategy for eliciting LLM reasoning is through model training or instruction tuning using a substantial amount of chain-of-thought (CoT) reasoning data (Chung et al., 2022; Cobbe et al., 2021b; Ling et al., 2017; Nye et al., 2021).

Prompting techniques, while effective, often encode task-specific human priors, thereby making it difficult to assess a language model's intrinsic reasoning abilities. Ideally, a language model should be able to reason independently and provide the optimal response, without requiring humans to tweak the prompts or refine repeatedly if the initial response is unsatisfactory. Model-tuning can be expensive and requires a substantial amount of supervised data. In this work, we explore a different perspective and ask: *Can LLMs reason effectively without prompting? And to what extent can they reason?* We find that, perhaps surprisingly, there exists a *task-agnostic* way to elicit CoT reasoning from pre-trained LLMs by simply altering the *decoding* procedure. Figure 1 illustrates this phenomenon: given a reasoning question, the LLM generates a wrong answer via the standard greedy decoding path, yet alternative top-k token inspection unveiled inherent CoT paths (e.g., decoding paths 2 and 4), which accurately resolved the query. This decoding modification bypasses prompting and is entirely unsupervised without the need for model tuning.

In more details, we formulate the input using the standard question-answer (QA) format: "Q:

© 2024 Google DeepMind. All rights reserved

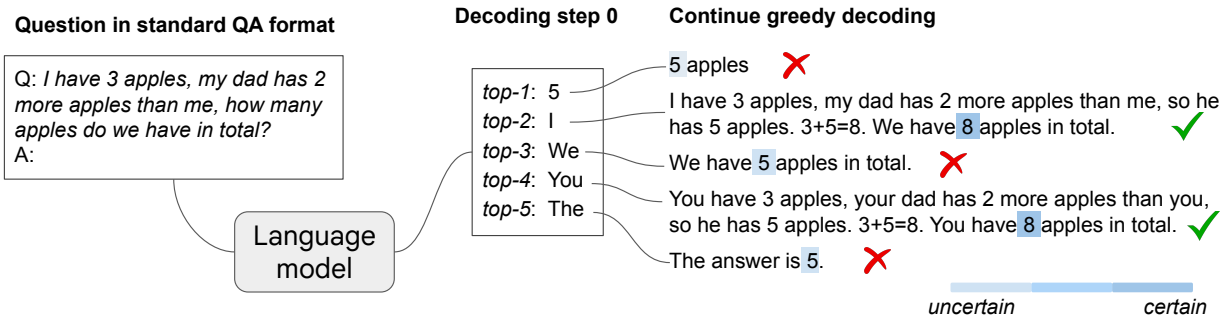

Figure 1 | **Illustration of CoT-decoding**. Pre-trained LLMs are capable of inherent reasoning without prompting by considering alternative top-k tokens, rather than solely relying on the top-1 greedy decoding path. Moreover, these models tend to display higher confidence in decoding the final answer (indicated by a darker shaded color) when a CoT reasoning path is present.

[question]\nA:".[1] While most existing work suggest that LLMs falter in such direct-QA scenarios on reasoning (Cobbe et al., 2021a; Kojima et al., 2022; Nye et al., 2021; Wei et al., 2022), our findings reveal a nuanced picture. We observe that LLMs indeed struggle with reasoning when relying solely on greedily decoded paths. However, when we consider alternative paths among the top-k tokens, CoT reasoning patterns *emerge naturally* within the decoding trajectories of LLMs. In addition, we have observed an interesting pattern: the model demonstrates increased confidence in the final answer when a CoT reasoning path is present in the decoding process. As illustrated in Figure 1, this is evident where paths 2 and 4 show heightened certainty in arriving at the correct answer "8", contrasting sharply with the high uncertainty in paths that lead to the incorrect "5". Leveraging this phenomenon, we develop a method to sift through the top-k decoding paths, which we refer to as **CoT-decoding**, thereby isolating the most reliable paths for model output.

Our contributions are summarized as follows:

- **We present a novel finding that LLMs can reason by simple decoding changes, without the use of prompting**. In contrast to prior research that focuses on refining prompts to elicit reasoning from LLMs, our work, for the first time, shows that the reasoning process can be readily elicited by simple decoding changes. Moreover, we challenge the prevailing notion in the literature that LLMs are inherently incapable of effective reasoning without prompting. We show that this belief is an artifact of considering only the greedy path during decoding, and the model's reasoning paths can be revealed by traversing the alternative decoding paths.
- **Our method enables a better understanding of LLMs' intrinsic reasoning capabilities without imposing human priors**. The employment of intricate prompting techniques often introduces various human priors, making it difficult to distinguish between the extent of "human teaching" and the degree to which LLMs can reason independently. Our approach bypasses the confounders introduced by prompting, enabling a more truthful assessment of the models' intrinsic reasoning abilities. Our study reveals that pre-trained language models *inherently possess reasoning capabilities* for many tasks including math and commonsense reasoning, and existing prompting approaches mostly serve the role of bringing those inherent reasoning paths forward as the top decoding paths. In contrast, the CoT-paths are less prevalent in complex and highly synthetic tasks, where the few-shot CoT demonstrations play a "teaching" role in guiding how models solve a task, with models primarily mimicing the format of these prompts to generate accurate reasoning paths.

[1]The QA format is only needed because without it a pre-trained language model will continue the question instead of answering. It is also the most basic formatting employed in existing works for pre-trained models.

- **We further propose CoT-decoding that reliably selects CoT-paths based on answer confidence.** We find that the language model's confidence in its final answers increases when a CoT is present in its decoding path. Leveraging this increased confidence, we propose **CoT-decoding** to select more reliable decoding paths, demonstrating significant improvements over greedy decoding across various reasoning benchmarks.

2. Chain-of-Thought (CoT) Decoding

2.1. Pre-trained Language Models Can Reason without Prompting

We investigate whether pre-trained language models inherently possess reasoning capabilities, without explicit prompts or human intervention. In Table 1, we show example decoding paths across math (GSM8K, Cobbe et al. (2021a)) and commonsense reasoning (year parity, Allen-Zhu and Li (2023)). We employ the *pre-trained* PaLM-2 large model (Anil et al., 2023) to compare its greedy decoding path ($k = 0$), predominantly used in state-of-the-art LLMs for reasoning tasks, with alternative decoding paths ($k > 0$), where k represents the choice of the k-th token at the *first decoding step*.

[GSM8K] *Kylar went to the store to buy glasses for his new apartment. One glass costs $5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?* **Greedy path**: $k = 0$: $**60.00** (0.029) **Alternative top-k paths**: $k = 1$: **60** (0.058) $k = 2$: Kylar needs to pay $**60** for 16 glasses. (0.058) ... $k = 7$: If Kylar buys 16 glasses, he will pay $**60**. (0.032) $k = 9$: We can calculate the price of 16 glasses by multiplying the price of one glass by 16. However, we need to remember that every second glass costs only 60% of the price. Therefore, we need to multiply the price of one glass by 16 and then subtract 40% of the price of 8 glasses. 16 x 5 = 80 8 x 5 = 40 40 x 0.4 = 16 80 – 16 = 64 Kylar needs to pay $**64** for 16 glasses. (0.994)
[Year Parity] *Was Nicolas Cage born in an even or odd year?* **Greedy path**: $k = 0$: Nicolas Cage was born in an **odd** year. (0.117) **Alternative top-k paths**: $k = 1$: **Even** (0.207) $k = 2$: **Odd** (0.198) $k = 3$: 1964, an **even** year. (0.949) $k = 4$: He was born in an **even** year. (0.0) ... $k = 7$: Cage was born in 1964, an **even** year. (0.978)

Table 1 | Examples of greedy decoded paths and alternative top-k paths over the PaLM-2 Large model. The model's confidence over the answers (bolded) are highlighted in blue (See §2.2 for details).

LLMs indeed cannot reason if we only consider the greedy decoding path. First, we observe that models employing greedy decoding often does not contain a CoT path, opting to solve problems directly. This tendency may stem from the model's skewed perception of problem difficulty, shaped by its pre-training on predominantly simpler questions. Consequently, the model is predisposed to immediate problem-solving. This observation aligns with findings in (Cobbe et al., 2021a; Kojima et al., 2022; Nye et al., 2021; Wei et al., 2022), which show that direct-answer prompts generally result in low accuracy on reasoning tasks even for large language models.